

CONFIDENTIAL INDUSTRIAL TELEMETRY REPORT

Turbine Unit 4 Vibration & Thermal Inspection • SIH26117 Test Standard

Facility: Sovereign Heavy Engineering - Sector 7
Date of Audit: September 2026
Governing SOP: SOP-IND-702 (Turbine Safety Limits) & ISO-10816
Lead Auditor: Senior Reliability Specialist

Subsystem / Component	Observed Reading	SOP Threshold	Deviation	Status
Turbine Bearing #1	71.2 deg C	<= 75.0 deg C	-3.8 deg C	COMPLIANT
Turbine Bearing #2	73.5 deg C	<= 75.0 deg C	-1.5 deg C	COMPLIANT
Turbine Bearing #3	88.4 deg C	<= 75.0 deg C	+13.4 deg C	CRITICAL EXCEEDANCE
Vibration Sensor V-2 (Radial)	4.8 mm/s	<= 3.0 mm/s	+1.8 mm/s	HIGH WARNING
Cooling Loop Delta T	6.8 deg C	>= 12.0 deg C	-5.2 deg C	RESTRICTED FLOW
Lubricant Viscosity (40C)	42 cSt	46 cSt +/- 2	-4 cSt	DEGRADED

Executive Audit Finding & Remediation Mandate:
Bearing #3 telemetry exhibits acute thermal anomaly (+13.4 deg C above SOP-IND-702 maximum permissible limit of 75.0 deg C), compounded by radial vibration exceeding ISO-10816 Zone B boundaries. Level 1 emergency isolation is mandatory within 12 hours to prevent catastrophic seizure.